Method of Procedure: Infrastructure Upgrade

Date: Tonight **Window:** 22:00 – 01:00 **PIC:** Linus & eAUTO Team

This document outlines the scheduled maintenance to upgrade the FortiGate High Availability (HA) cluster and two Synology NAS devices. Please review the specific actions required prior to the maintenance window.

> **CRITICAL ACTION REQUIRED: eAUTO Team (Pre-Maintenance)**
>
> The Synology NAS
> **172.30.202.38**
> will undergo a full system reboot during the upgrade. Because an eAUTO VM utilizes an active NFS link to this NAS, the eAUTO team
> **must manually disable the dependent VM services and safely unmount the NFS datastore/share**
> before 22:00. This is to prevent stale file handles, application freezing, or data corruption during the storage interruption.

1. Execution Schedule

Time	Phase	Activities	PIC
22:00 – 22:30	Pre-Checks & Prep	• Take config backups (FortiGate & NAS DSS). • Verify pre-downloaded images and checksums. • **eAUTO: Disable services and unmount NFS on .38.**	Linus / eAUTO
22:30 – 22:50	FortiGate HA Upgrade	• Upgrade secondary node to v7.4.13. • Trigger cluster failover; upgrade primary node. • Verify session sync and routing stability.	Linus
22:50 – 23:25	NAS 1 (.38) Upgrade	• Upgrade 7.2.2-72806 to DSM 7.3 LTS (2-step path) & reboot. • Verify volume health and network services.	Linus
23:25 – 00:00	NAS 2 (.40) Upgrade	• Upgrade 7.1.1-42962 to DSM 7.3 LTS (sequential path) & reboot. • Verify volume health and network services.	Linus

Time	Phase	Activities	PIC
00:00 – 01:00	Testing & Sign-off	• **eAUTO: Remount NFS and enable VM services.** • Joint verification of storage access and app health. • Monitor logs and close maintenance window.	Linus / eAUTO

2. Downtime & Impact Analysis

FortiGate Firewall Zero network blackout. The HA Active-Passive pair will seamlessly failover. There will be a momentary session disruption (5–15 seconds), during which VPN tunnels and active TCP sessions will automatically re-establish.

Synology NAS .38 Approximately 20–25 minutes of downtime. All SMB/CIFS, NFS, and iSCSI services will be unreachable during the dual reboot cycle. *(Mitigated by pre-disabling eAUTO NFS links).*

Synology NAS .40 Approximately 20–25 minutes of downtime (performed sequentially after .38 is fully online).

3. Testing & Verification (Post-Upgrade)

- **Network (Linus):** Verify FortiGate HA synchronization status, VPN site-to-site tunnels, and outbound routing.
- **Storage (Linus):** Confirm both NAS units report DSM 7.3 LTS. Check Storage Manager to ensure all pools/volumes are healthy.
- **Application (eAUTO):**
  - Successfully remount the NFS link on the VM pointing to NAS .38.
  - Enable application services and verify file read/write capabilities on both .38 and .40.
  - Confirm database backup targets are accessible.

4. Rollback Plan

The cut-off time for a Go / No-Go decision is **23:45**. If systems exhibit critical failures, the rollback procedure will be executed:

- **FortiGate:** In the event of synchronization failure or heavy packet drop, manually fail back to the surviving v7.4.12 node, re-flash the problematic unit, and restore the pre-maintenance `.conf` backup. (Estimated time: 15 mins).
- **Synology NAS:** If the firmware fails to initialize or a volume crashes post-update, perform a Mode 2 System Reset (reinstalls base DSM without touching data partitions) and restore the saved `.dss` configuration. (Estimated time: 30–40 mins).